DEVOPS OPEN-SOURCE PROJECT WORKBOOK

Make GitHub Actions Checks Independently Addressable

A practical contribution workbook for Kubernetes SIG Testing issue #389



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Portfolio repository: github.com/Chiranjibee18/AWS-VPC-Peering-Project
Target issue: kubernetes-sigs/prow #389 | Status must be re-checked before coding

LEARN. BUILD. TEST. EXPLAIN.

01 / PROJECT BRIEF

Understand the problem

The issue reports that several GitHub Actions workflows publish the same check name: **Run on Ubuntu**. GitHub branch protection and rulesets use check/status contexts to identify required checks. If multiple jobs share the same context, a repository may be unable to require only the intended workflow check.

The goal

Give the relevant workflow jobs unambiguous names so downstream repositories can require E2E Tests, Tests, Lint, or Govulncheck independently—while keeping pull-request triggers and unrelated workflow behavior unchanged.

> **Important status check: the issue text mentions PR #432 and an earlier PR #392. Verify whether either change has merged and whether issue #389 is still actionable before writing code.**

Problem → change → outcome

Area	Before	Desired outcome
Check identity	Multiple workflows report “Run on Ubuntu”.	Each relevant check has a distinct, stable name.
Branch protection	A required context may match multiple jobs.	Maintainers can select the check they actually need.
Workflow behavior	Existing pull-request workflows run today.	Triggers, commands and test coverage remain unchanged.
Downstream repos	Synced workflows may expose ambiguous contexts.	Repositories can configure required checks independently.

Scope and non-goals

In scope: inspect workflow YAML, identify duplicate job/check names, make the smallest naming change, and validate the resulting check contexts. Out of scope unless maintainers request it: redesigning CI, changing test commands, changing workflow triggers, or modifying unrelated jobs.

02 / INVESTIGATE THE REPOSITORY

Inspect before editing

Start from the issue and pull requests, not from assumptions. The issue may have been resolved already, and the exact intended naming convention must come from the repository and maintainers.

Links to review

Resource	Why it matters
Issue #389 — original request	Confirm current status, latest maintainer comments and expected behavior.
PR #432 — linked change	Check whether the proposed fix is open, merged, closed or superseded.
PR #392 — earlier change	Read the changes and discussion referenced by maintainers.
Repository CONTRIBUTING.md	Follow local setup, testing, sign-off and pull-request requirements.

Repository navigation

```
# Use your fork URL after forking the upstream repository
git clone https://github.com/YOUR-USERNAME/prow.git
cd prow

# Review contributor guidance and workflow files
find . -name 'CONTRIBUTING.md' -o -path '*/.github/workflows/*.yaml' -o -path '*/.github/workflows/*.yml'

# Search for the duplicate job name
grep -RIn --include='*.yml' --include='*.yaml' 'Run on Ubuntu' .github/ . 2>/dev/null

# Inspect local changes before editing
git status
git log -5 --oneline
```

Repository note: your existing portfolio repository is AWS-VPC-Peering-Project. It demonstrates cloud networking; it is not the Kubernetes Prow source repository. Keep the contribution in a fork of the actual upstream project.

Before you code

- Confirm the issue is still open and needed.
- Check the status and diff of #432 and #392.
- Read CONTRIBUTING.md and relevant workflow files.
- Ask a maintainer if the scope or naming convention is unclear.
- Record the upstream branch and current commit.

Make the smallest safe change

The likely implementation is a small change to job names in the affected GitHub Actions workflow YAML files. The exact files and final names must be determined from the current upstream code and maintainer guidance; this workbook does not claim that a fix has already been implemented.

Typical Git workflow

```
# Create a focused branch
git switch -c fix/unique-github-check-names

# Edit only the confirmed workflow files, then inspect the diff
git diff --check
git diff
git status

# Stage only the files you changed
git add path/to/confirmed-workflow.yaml
git commit -m "ci: give workflow checks unique names"

# Push your branch to your fork
git push -u origin fix/unique-github-check-names
```

Conceptual YAML example

This illustrates the idea only. Do not copy it blindly; match the repository's actual workflow structure and naming conventions.

```
jobs:
  e2e:
    name: E2E Tests
    # Preserve existing runs-on, steps and conditions

  tests:
    name: Tests
    # Preserve existing runs-on, steps and conditions

  lint:
    name: Lint
    # Preserve existing runs-on, steps and conditions

  govulncheck:
    name: Govulncheck
    # Preserve existing runs-on, steps and conditions
```

Review checklist

Check	Pass criteria
Names	Each affected check context is distinct and descriptive.
Scope	Only intended workflow/job names change.
Behavior	Triggers, runners, conditions, commands and dependencies are unchanged unless explicitly required.
Diff quality	git diff --check passes; no unrelated files or formatting churn.
Compatibility	Names are stable enough for downstream branch protection configuration.

Prove the change works

A successful YAML edit is not enough. Confirm the resulting check names on a test pull request or in the repository's supported CI validation, and verify that unrelated behavior did not change.

Test	How to validate	Expected result
YAML / style	Run the repository's documented validation commands; inspect `git diff --check`.	Workflow files parse and style checks pass.
Check-name uniqueness	Inspect the check list for a PR after the workflows run.	The relevant jobs show distinct check contexts.
Pull-request behavior	Compare triggers, conditions and job steps with the original workflow.	The same workflows still run for the same PR events.
Independent selection	In a safe test repository or with maintainer guidance, require only one of the new contexts.	The selected check can be required without requiring the others.
Downstream compatibility	Ask maintainers how synced repositories consume the workflow names.	The change supports independent branch protection configuration.

Common failure modes

Symptom	What to inspect
Check still appears duplicated	Check whether another job or reusable workflow still publishes the old name.
Required check is pending forever	Confirm the exact context name and whether the job runs for the target event.
Workflow no longer runs	Review `on`, `if`, path filters and reusable workflow inputs; avoid unrelated edits.
CI passes but check name is wrong	Inspect the displayed check context, not just the workflow file or workflow title.
Maintainer says work is duplicate	Stop and coordinate; do not submit a redundant PR.

Evidence rule: record actual results only. If CI has not run or a PR has not been reviewed, label it “not yet verified” rather than “passed” or “merged”.

05 / SUBMISSION & CAREER EVIDENCE

Turn the contribution into a portfolio story

A strong open-source contribution is not just a changed line of YAML. It is a clear problem statement, a minimal patch, evidence that behavior is preserved, and a professional review process.

Pull request description outline

```
## What problem does this solve?
Several jobs expose the same check context, making independent
branch-protection configuration difficult.

## What changed?
Describe the exact workflow/job names changed and why.

## Validation
- [ ] Repository-required checks completed
- [ ] Confirmed unique check contexts
- [ ] Confirmed existing PR triggers and behavior are unchanged

## Related issue
Fixes #389
```

Evidence checklist

- Link to the upstream issue and the PR you actually submitted.
- Screenshot or text evidence of the final distinct check names.
- CI run link and status, if available.
- Short summary of review feedback and any revisions.
- Clearly state whether the PR is open, merged, closed or still awaiting review.

Interview questions to prepare

Question	Key idea to explain
What is a GitHub Actions workflow vs. job?	A workflow is the automation definition; jobs are units of work within it.
What is a check context?	The name associated with a reported status/check that branch protection can require.
Why can duplicate names be a problem?	A required status context can be ambiguous when multiple jobs report the same name.
How did you avoid regressions?	Kept the patch minimal and verified triggers, conditions, steps and resulting check names.
What if a PR already fixes the issue?	Confirm with maintainers and find another task rather than duplicating work.

Your existing portfolio

AWS VPC Peering Project — github.com/Chiranjibee18/AWS-VPC-Peering-Project

Use this repository as evidence of your AWS networking practice. Add the Kubernetes contribution as a separate portfolio entry only after you have a real issue/PR link; do not imply that the AWS repository contains the Kubernetes fix.

Recommended next action: inspect issue #389 and PRs #432 / #392 first. If the issue is still actionable, confirm scope with a maintainer, then implement and test a minimal change.